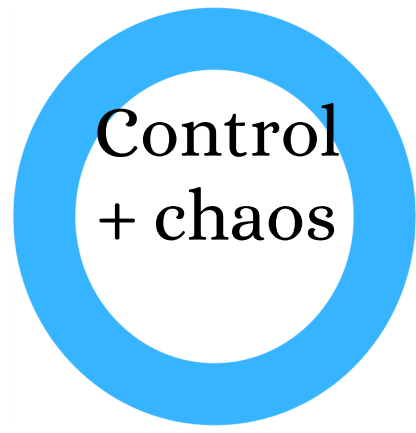


TITLE PAGE

- **Problem Statement Title-** Lack of personalized, complete, and motivating fitness guidance
- **Theme :** Fitness / Healthtech
- **PS Category:** Software
- **Team Name :** Control + chaos

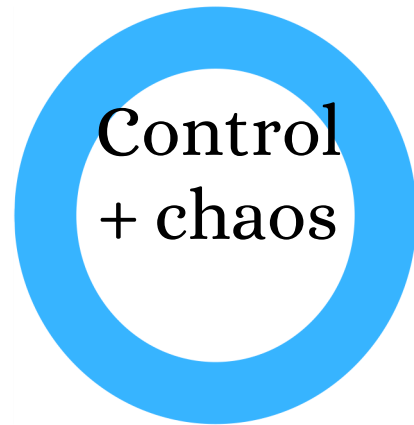


IDEA TITLE



WORKFIT

~ AI-Powered Digital Fitness & Indian Nutrition Companion ~



TECHNICAL APPROACH



- **Multi-Modal AI Perception Pipeline:** Real-time visual analysis for automated meal itemization, portion estimation, and posture biomechanics.
- **Dual Grounded Nutrition Engine:** On-device SQLite FTS5 database (~8,000 foods) integrating USDA & Indian Nutrient Databank (INDB) for authentic desi meals.
- **Deterministic Safety & Adaptive Math:** Mifflin-St Jeor & EWMA trend algorithms with hard clinical minimum safety floors (1,200/1,500 kcal) preventing extreme deficits.
- **Edge-First Standalone Mobile Architecture:** Offline-first React Native + native Android architecture with local data ownership and private AI gateway connectivity.
- **End-to-End User Flow:** Onboarding & Goal Anchoring → Computer Vision Body Scan → Multi-modal Food Perception → Adaptive Targets → Streak Reminders.

Feasibility & Viability:

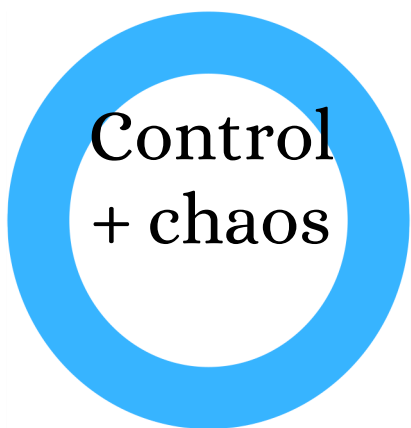
- **Zero Dedicated Hardware:** Runs directly on standard smartphone cameras; no expensive wearables or DEXA scans required.
- **Offline-First Core:** Database lookups, calorie calculation, hydration, and workouts run 100% locally on-device without cloud lock-in.

Potential Challenges and Risks:

- **Perception Accuracy:** Estimating portion volume and invisible cooking fats (ghee/oil) in complex Indian dishes.
- **LLM Hallucination Risk:** Unchecked generative AI suggesting unsafe, non-clinically backed crash diets.

Mitigation & Safety Strategies:

- **Cooking Fat Assumption Models:** Resolver bias filters and default cooking fat assumption fillers calibrated to Indian food preparation.
- **Deterministic Safety Guardrails:** Hard-coded clinical minimum calorie floor clamps and validated biomechanical joint angle checks.
- **Local Data Ownership:** All health, weight, and scan records stored securely on-device with zero mandatory cloud exposure.



IMPACT AND BENEFITS



Potential Impact on Target Audience:

- **Democratizes Fitness & Nutrition:** Provides 24/7 personalized coaching for students, beginners, and professionals who cannot afford costly personal trainers.
- **Engineered for Indian Dietary Patterns:** Solves the national pain point of tracking authentic Indian home-cooked meals, thalis, and regional staple foods.

Key Benefits of the Solution:

- **Biomechanical & Musculoskeletal Health:** AI posture scans identify forward head posture, shoulder asymmetry, and pelvic tilt, prescribing targeted corrective routines.
- **Habit Formation & Engagement:** Gamification via streak milestones, hydration tracking, and interactive goal recalibration keeps users consistently motivated.
- **Economic Accessibility:** 100% free and open-access edge architecture compared to commercial alternatives charging ₹3,000–₹15,000/year subscription fees.

1) Scientific Foundations & Verified Databanks:

- **Indian Nutrient Databank (INDB):** Macro & micronutrient dataset for regional Indian recipes (Harvard / Lindsay Jaacks).
- **USDA FoodData Central (FDC):** Standard Reference and Branded Food database foundation.
- **Mifflin-St Jeor & EWMA Algorithms:** Clinically validated metabolic rate calculation & Hacker's Diet trend smoothing.

2) Competitor Flaws vs. WorkFit AI Solutions:

- **Misses Indian / Desi Food:** Competitors fail on regional dishes → WorkFit integrates INDB with regional portion priors.
- **Hidden Ghee, Oil & Sugar:** Often missed in food photos → WorkFit deploys culinary cooking-fat assumption fillers.
- **Noisy Weight Spikes & False Precision:** Raw scale fluctuations mislead users → WorkFit uses EWMA trend smoothing.
- **Hallucinated Crash Diets:** Unsafe AI diet plans → WorkFit enforces deterministic clinical calorie & macro safety clamps.
- **Subscription Paywalls:** AI tools locked behind ₹3,000–₹15,000/yr → WorkFit is 100% edge-first & accessible to all.